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St. Joseph's College of Arts & Science (Autonomous)
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CH510S – INORGANIC CHEMISTRY - III

Time : 3 hrs **Max Marks :75**

SECTION – A (10X1=10)

Answer **ALL** Questions

I. Choose the correct answer

1. The process of removal of gangue from the ore
   a) Ore-dressing b) Metallurgy
   c) Calcination d) Smelting

2. Which one of the following ions has the maximum magnetic moment?
   a) T_i^{3+} b) Co^{2+} c) V^{3+} d) Cr^{3+}

3. The number of ligands attached to the central atom
   a) Ligands b) Co-ordination number
   c) Central metal atom d) Isomerism

4. $[Co(NH_3)_5\ Br]So_4$ and $[Co(NH_3)_5\ So_4]\ Br$ Show
   a) Hydrate isomerism b) Linkageisomerism
   c) Ligand isomerism d) Ionization isomerism

5. Which among the following will be paramagnetic?
   a) $Cr(CO)_6$ b) $Fe(CO)_5$ c) $V(CO)_6$ d) $Fe_2(CO)_9$

6. The structure of $Ni(CO)_4$ is
   a) Square planer b) Linear c) Tetrahedral d) Octahedral

7. The ion which is expected to have Jahn-Teller distortion in an octahedral field is
   a) Co^{2+} C strong field b) Co^{2+} (low-spin)
   c) Ni^{2+} d) Mn^{2+} (Weak field)

8. Which of the following ions has Zero CFSE?
   a) Fe^{2+} b) Zn^{2+} c) Co^{3+} D) Ni^{2+}

9. Breagg's equation is given by the expressions
   a)$n\lambda = 2d \sin\theta$ b)$n\lambda = d\sin\theta$
   c)$n\lambda = \frac{d \sin\theta}{2}$ d)$n\lambda = d \sin \theta/2$

10. Silicon doped with indium is an example for
    a) Intrinsic semiconcuctor b) P-type semiconductor
    c) n-type semiconductor d) All the above

SECTION – B (10X2=20)

Answer any **TEN** Questions

11. What is meant by metallurgy?

12. What are transition elements?

13. Name the following according to IUPAC tules.
    i) $[Ni(CN)_4]^{2-}$ ii) $[Cu(NH_3)_4]^{2+}$

14. What is Effective Atomic Number (EAN) rule?

15. Which one, Cu^{2+} or Zn^{2+}, is diamagnetic and why?

16. List out the failures of VBT.

17. Write down the structure and hybridization of $[Ni(CO)_4]$.

18. Define the term Trans effect with an example.

19. Write briefly note on semiconductor.

20. What is Bragg's equation?

21. Give any two preparation of V_2o_5.

22. What are ligands in coordination compounds?

23. Define the following with examples
    i) High spin complexes ii) Low spin complexes

24. $[Fe\ F_6]^{3-}$ is colourless but $[Fe\ cl_4]^-$ is yellow coloured. Why?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

25. i) How is $(NH_4)_2$ Mo O_4 prepared? Write down its equation. **(3)**
    ii) Explain froath floatation process. **(4)**

26. i) Explain briefly Geometrical isomerism in tetrahedral complexes. **(2)**
    ii) How are ligands classified? **(5)**

27. i) Write a note on the concept of primary and secondary valencies of elements advanced by Werner. **(5)**
    ii) Calculate the EAN of Cr(III) in $[Cr(CN)_6]^{3-}$. **(2)**

28. i) Distinguish between VBT and CFT in coordination complex. **(4)**
    ii) Explain the hybridization and structures of $Fe(CO)_5$. **(3)**

29. i) What are crystal defects? How are they classified? **(2+2)**
    ii) Write a note on super conductors with an example. **(3)**

30. i) How will you distinguish between the following pairs of isomers. $[Cr(NH_3)_3(No_2)_3]$ and $[Cr(NH_3)_6]$ $[Cr(No_2)_6]$. **(4)**
    ii) Write a note on various oxidation states of Mn. **(3)**

31. i) Explain the purification of Ti using Van Arkel method. **(3)**
    ii) Write a note on Jahn-Teller effect. **(4)**

32. i) Explain the Band theory of metals. **(3)**
    ii) Which one, Cu^+ or Cu^{2+}, is colourless and why? And write their electronic configuration. **(4)**

SECTION - D (1X10=10)

Answer any **ONE** Question

33. Write a note on splitting of 'd' orbitals in octahedral, distorted octahedral, square planar and tetrahedral environments. **(10)**

34. i) X-rays of λ=0.134nm gave a first order diffraction from the surface of a crystal when the value of θ=10.5^o Calculate the distance between the plane in the crystal parallel to the surface examined. **(5)**

    ii) The inter planar distance in a crystal used for X-ray diffraction is 2°A (0.2nm) the angle of incidence (θ) of X-rays is 9.0^o when the first order diffraction is observed. Calculate the wave length of the X-rays. **(5)**

35. i) The EAN of each Mn (atomic number 25) in its carbonyl $Mn_2(CO)_{10}$ is 36. What will be the structure of the carbonyl? **(3)**

    ii) Explain, which one, Fe^{2+} or Fe^{3+}, has higher magnetic moment and why? **(3)**

    iii) Write a note on F-centre. **(4)**

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